



Cell Biology

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*Type and functions of organic and
inorganic substances*

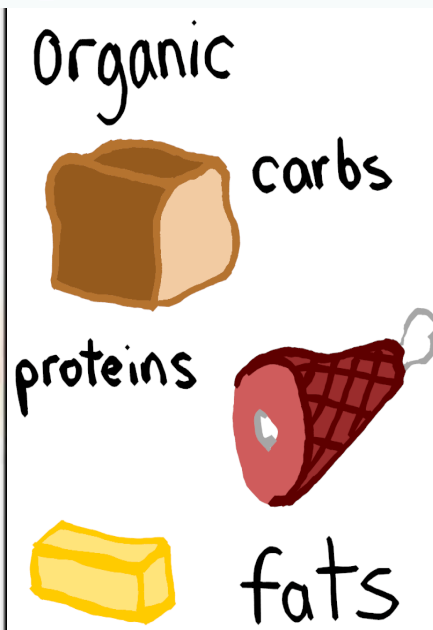
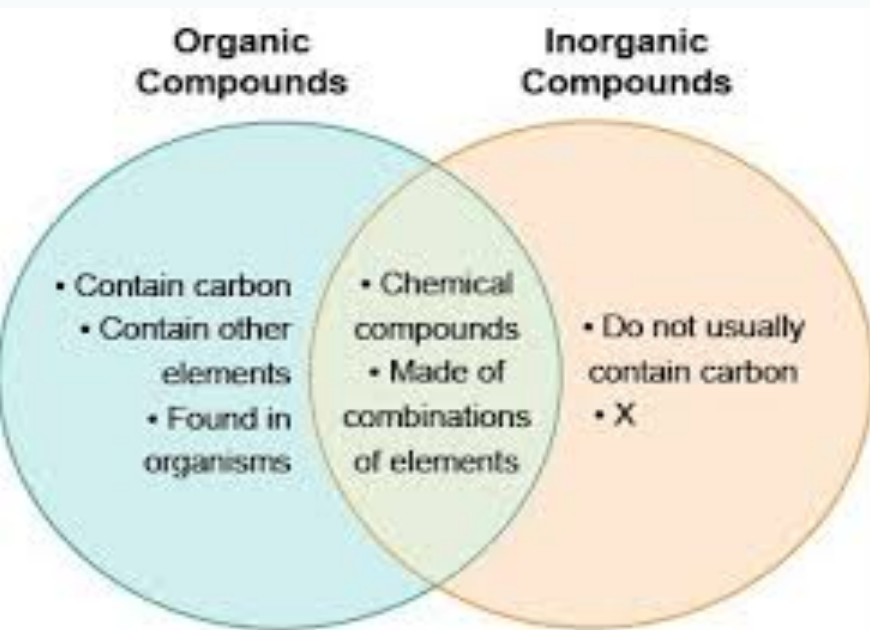
Inorganic components of the cell

Inorganic components are found in cells either in free ionized state, or bounded with other organic substances

1- Ions : ex(Sodium (Na), Potassium (K), Calcium (Ca), Magnesium (Mg), Chloride (CL), Nitrate (NO₃)).

2- Salts : ex (NaCl).

3-Trace elements: are found in a very small amounts:
ex: Molybdenum (Mo⁺⁴, Mo ⁺³) & Cobalt (Co⁺²).



Cell is composed of...

Organic compounds

Carbohydrates

Lipids

Proteins

Nucleic Acids

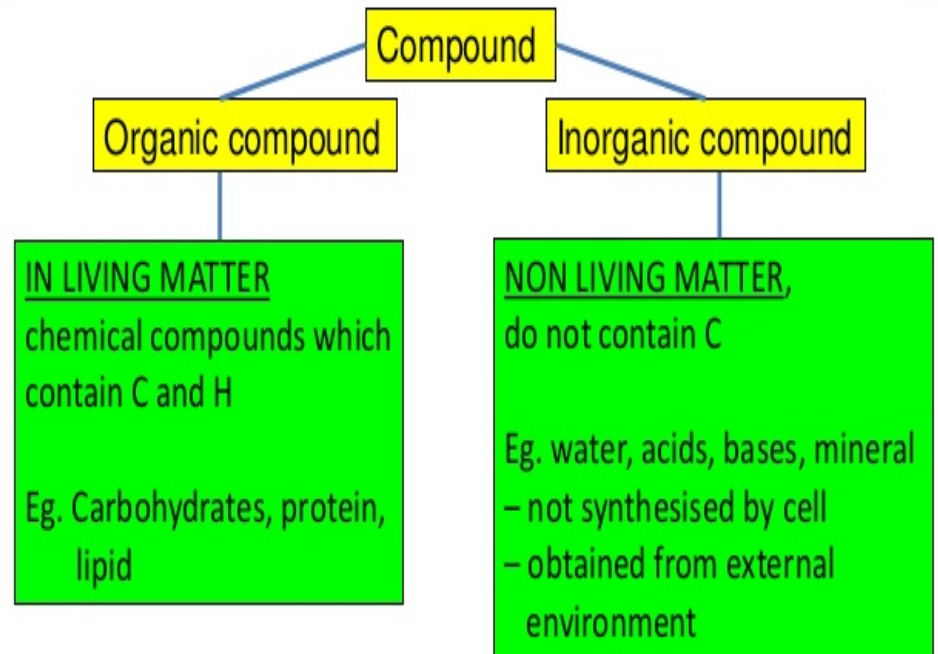
Inorganic compounds

Water 90-95%

Salts e.g., NaCl

Elements K, Ca, Fe, Na

Gases CO₂ and O₂



- **1-Calcium (Ca)**
- **Food source:** most fruit and vegetables, meat, dairy products
- **Main functions:** strong bones and teeth; muscle contraction; blood clotting; nerve function
- **Deficiency Disease:** **rickets, osteoporosis**
- **2-Phosphorus (P)**
- **Food source:** nuts, meat, dairy products
- **Main functions:** strong bones and teeth; nerve function; part of nucleic acids and cell membranes
- **Deficiency Disease:** **rickets, osteoporosis**
- **3-Magnesium (Mg)**
- **Food source:** nuts, meat, dairy products
- **Main functions:** strong bones and teeth; nerve and muscle function; energy production
- **Deficiency Disease:** **osteoporosis, muscle cramps**

- **4-Potassium (K)**
- **Food source:** bananas, meat, dairy products
- **Main functions:** growth and maintenance, water balance, heart function
- **Deficiency Disease:** **muscle cramps; heart, kidney and lung failure**
- **5-Sodium (Na)**
- **Food source:** table salt, fruit and vegetables, onions, fresh fruits, sweet potato, broccoli, pumpkin seeds, eggs, milk .
- **Main functions:** regulates blood pressure and volume; muscle and nerve function
- **Deficiency Disease:** **muscle cramps**

6-Iron (Fe)

- **Food source:** meat, legumes
- **Main functions:** part of hemoglobin (the oxygen transport protein); part of some enzymes
- **Deficiency Disease:** **Anemia**

7-Iodine (I)

- **Food source:** seafood, iodated salt
- **Main functions:** production of hormones by the thyroid gland; strong bones and teeth; good hair; skin and nails
- **Deficiency Disease:** **goiter, stunted growth, mental problems**

8-Zinc (Zn)

- **Food source:** seafood, meat
- **Main functions:** immune function; male reproductive system
- **Deficiency Disease:** **stunted growth, prostate problems**

9- Cobalt

- **Food source:** fish. nuts. green leafy vegetables, such as broccoli and spinach.
- **Main functions:** Cobalt is an essential element and only a little over **1 mg** is present in an adult human, with the largest amounts being in liver, skeletal muscles, bone, hair, adipose tissue and blood. As **essential part of vitamin B₁₂**, is important during **creation of red blood cells**.
- **Deficiency Disease:** **Anemia, vitamin B-12 deficiency**

Cell Organic Components

Introduction

- From the origin of the first cell to current variety of organisms, carbon has played a prominent role in the evolution of life on earth.
- Although a single cell is composed of (70-90)%water, most of the rest consists of carbon based compounds, Proteins, Nucleic acids ,Carbohydrates, and other molecules that distinguish living matter from inanimate materials are all composed of carbon atoms bounded

Cell Organic Components

Introduction

to one another and to atoms of other elements, Hydrogen, oxygen, nitrogen, sulfur and phosphorous are other common ingredients of these compounds.

- Compounds containing carbon are said to be organic
- The percentages of the major elements of life { C, O, H, N, S and P } are quite uniform from individual to individual and even from species to species.

Cell Organic Components

Introduction

- Four major organic compounds (macromolecules) in cell are :
- **1- Carbohydrates**
- **2- Lipids**
- **3- Proteins**
- **4- Nucleic acids**

1- Carbohydrates

- Include sugar and their polymers.
- ***monosaccharides** The simplest carbohydrates are the single sugars glucose (which is the source for energy). Ribose (in DNA & RNA).
- ***Disaccharides** (double sugar) such
- maltose (two glucose),
- sucrose (glucose + fructose),
- lactose (glucose + galactose).

1- Carbohydrates

- **Polysaccharides**: polymers of few hundreds to few thousands of monosaccharides connected to each other by glycoside bonds.
- **Homopolysacchrides** are polymers containing only one kind of monosaccharides (glycogen, starch and cellulose are polymers of glucose)
Heteropolysaccharides are polymers containing more than one kind of monosaccharides.

1- Carbohydrates

- polysaccharides are either
- * **storage polysaccharides**
- hydrolyzed as needed to provide sugar for cells. Starch is a storage polysaccharide represent a storage energy for plants and glycogen is another polysaccharide which is stored by animals mainly in liver and muscle cells. Its hydrolyzed when the demand for sugar increase.

1- Carbohydrates

- * Structural polysaccharides
- Living organisms build strong materials from structural polysaccharides, for example **cellulose** is a major component of the plant's cell wall. The cellulose fibrils in our food pass through the digestive tract and are eliminated with the feces. Although, cellulose is not a nutrient for human it is an important part of a healthful diet.

Protein Introduction

- Constitute more than 50% of the dry weight of most cells. Proteins are used for structural support, storage, transport of other substances, signaling from one part of the organism to another, movement and defense against foreign substances. In addition, as enzymes, proteins selectively accelerate chemical reactions in the cell.
- A human has thousands of different kinds of proteins, each with a specific structure and function. All proteins are formed from the same set of 20 amino acids

Proteins are classified according to their function in human as follows

- **1-Structural protein.**
- **2-Storage protein.**
- **3-Transport protein.**
- **4-Hormonal protein.**
- **5-Receptor protein.**
- **6-Contractile protein.**
- **7-Defensive protein.**
- **8-Enzymatic protein.**

1-Structural proteins:

- These are for structural support, like
- *collagen and elastin that provide a fibrous framework in animal connective tissues, such as tendon and ligaments.
- *Keratin is the protein of hair, horns, feathers and other skin appendages of animals.

2-Storage protein

- These are used as source of amino acids, for example,
- *Casein, the protein of milk, is the major source of amino acids for baby mammals.

3-Transport proteins:

- Proteins that has role in transporting substances in human body, for example,
- * Hemoglobin, the iron containing protein of blood, transport oxygen from the lungs to other parts of the body.
- *Other proteins transport molecules across cell membrane

4-Hormonal proteins:

- These coordinate bodily activities, for example
- * Insulin hormone secreted by the pancreas, regulate the concentration of sugar in the blood.

5-Receptor proteins:

- These proteins help the cell to respond to chemical stimuli.
- *Receptor proteins built into the membrane of a nerve cell detect chemical signals released by other nerve cell.

6-Contractile proteins :

- Are responsible for tissue movement, for example,
- *Actin and myosin are responsible for the movement of muscles.
- *Other contractile proteins are responsible for the undulation of cilia and flagella which propel many cells.

7-Defensive proteins :

- Defense proteins such as antibodies, which are also known as immunoglobulin, are designed to be able to tell when a foreign body has entered the bloodstream.

8-Enzymatic proteins:

- These accelerate body chemical reactions. For example,
- *digestive enzymes hydrolyses the polymers in food.

Lipids

Introduction

They are chemically diverse compounds, grouped together because they share one important trait, **they have little or no affinity for water.** Lipids hydrophobic behavior is based on their molecular structure. The chemical properties of lipids make them ideal structural components of cell membrane.

Families of lipids

- a-Fats.

Saturated and unsaturated

- b-Phospholipids.

c-Steroids.

1-Fats

- Are large molecules constructed from two kinds of smaller molecules, **glycerol and fatty acids**.
- Fats are of two types,
- *saturated (animal fats),
- **unsaturated (vegetable fats).
- An example of fats is triglyceride.

1-Fats

- A diet rich in saturated fats is one of several factors that may contribute to the cardiovascular disease known as atherosclerosis, in this condition deposits called plaque develop on the internal lining of blood vessels impeding blood flow.

2- Phospholipids

- Are structurally related to fats, but they have only two fatty acids rather than three, the third carbon of glycerol is joined to a phosphate group.
- Because of structures, phospholipid are well situated to their function as a major constituents of cell membrane.

3- steroids

Are lipids characterized by a carbon skeleton consisting of four inter connected rings.

- An important steroids is **cholesterol**, a common component of the membranes of animal cells.
- Cholesterol is also the precursor from which most of other steroids are synthesized .

3- steroids

- For example, many hormones, including sex hormones of vertebrates, are steroids modified from cholesterol.
- Thus cholesterol has important functions in animals, although a high concentrations of cholesterol in the blood may contribute to **atherosclerosis**.

Biological functions of lipid

- 1- They provide an important form of energy storage.
- 2- Lipids are the major components of cells membrane.
- 3- Play important role in cell signaling.

Nucleic acids

Introduction

- All cells of living Organisms contain nucleic acids in the form of deoxy ribonucleic acid (**DNA**) & ribonucleic acid (**RNA**).
- **DNA** Is the major store of genetic information's this information is transmitted by transcription to RNA Molecules and these genetic materials are inherit from their parents.

Nucleic acids

Introduction

- DNA molecule is very long & consists of hundreds of thousand of genes, each occupying a specific position along the single molecule.
- When a cell reproduces it self, its DNA is copied and passed along from one generation of cell to the next, but The DNA is not directly involved in running the operation of the cell, proteins are required for the implementation of genetic information.

- Nucleic acids are composed of nucleotides and each of them is composed of three basic units:

1-nitrogen base (purin & pyrimidine).

2-sugar (pentose).

3-phosphoric acid

Nucleotides and Nucleosides

- Nucleotide =
 - Nitrogenous base
 - Pentose
 - Phosphate
- Nucleoside =
 - Nitrogenous base
 - Pentose
- Nucleobase =
 - Nitrogenous base

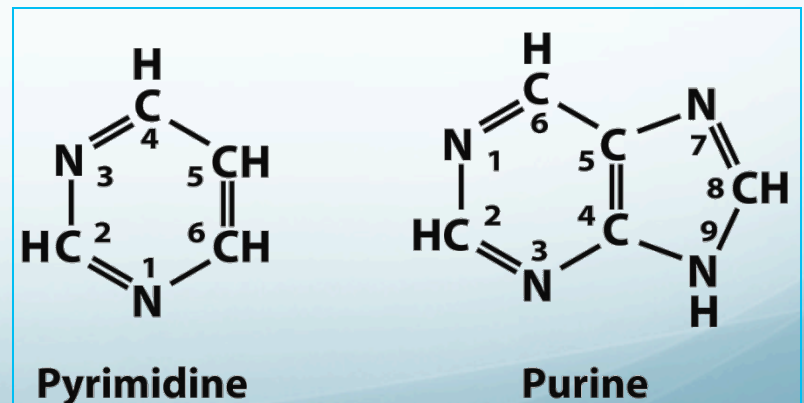
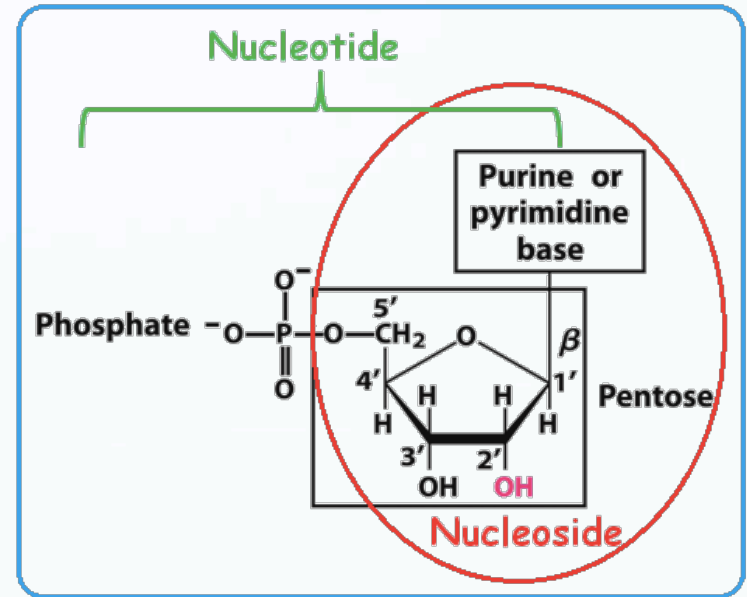
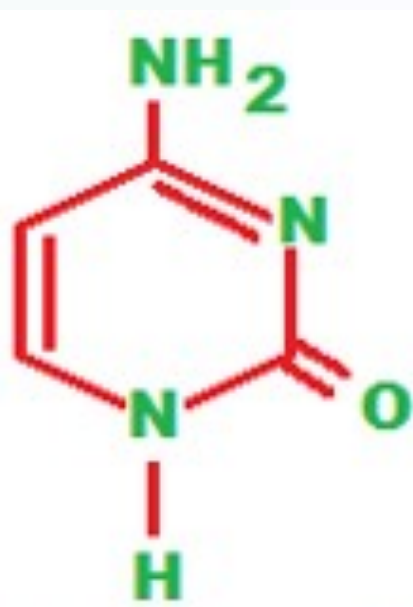


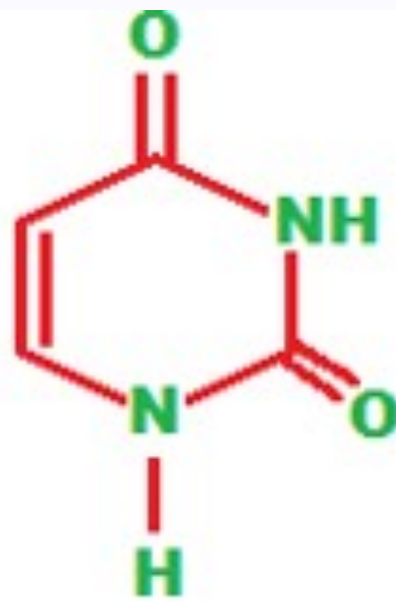
Figure 8-1b
Lehninger Principles of Biochemistry, Sixth Edition
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1-nitrogen base (purin & pyrimidine)

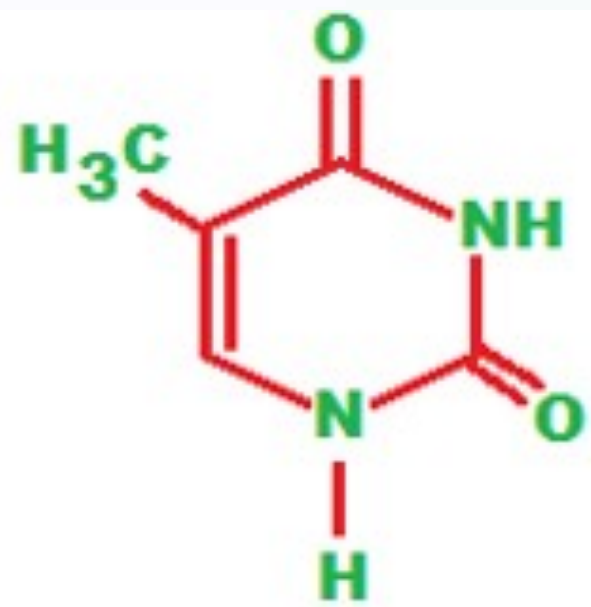
- **a-Pyrimidine** : is characterized by a Six –member ring
Made of carbon & nitrogen atoms .
- Members of pyrimidine family are nitrogenous bases
named
- **1-Cytosine (C) 2-Thymine (T) 3-Uracil (U).**
- Thymine is found only in DNA & Uracil only in RNA.



Cytosine (C)



Uracil (U)

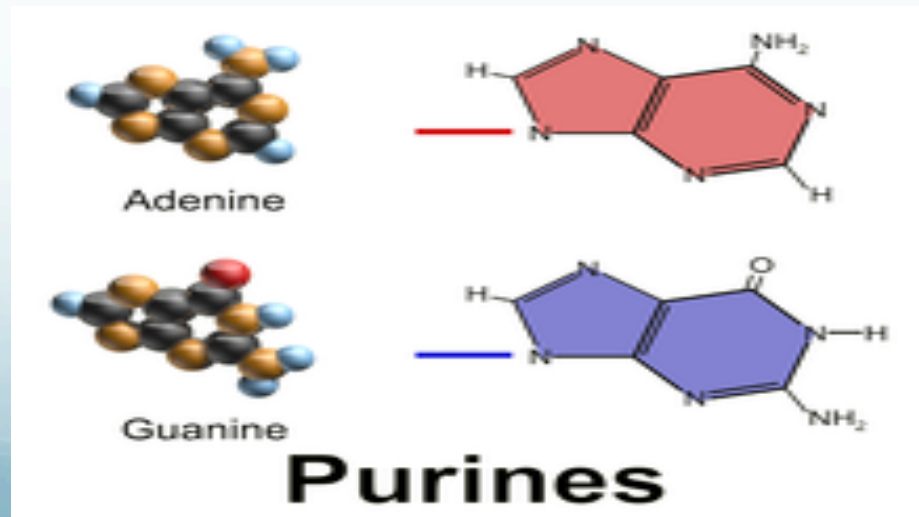


Thymine (T)

Pyrimidines

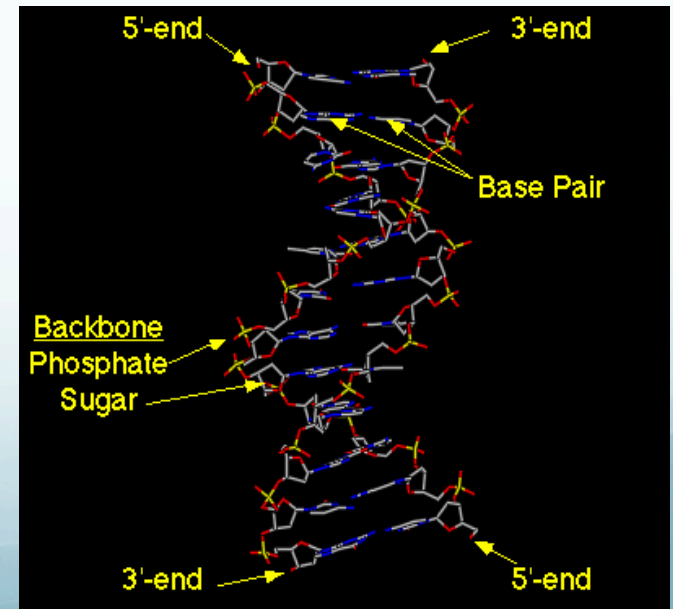
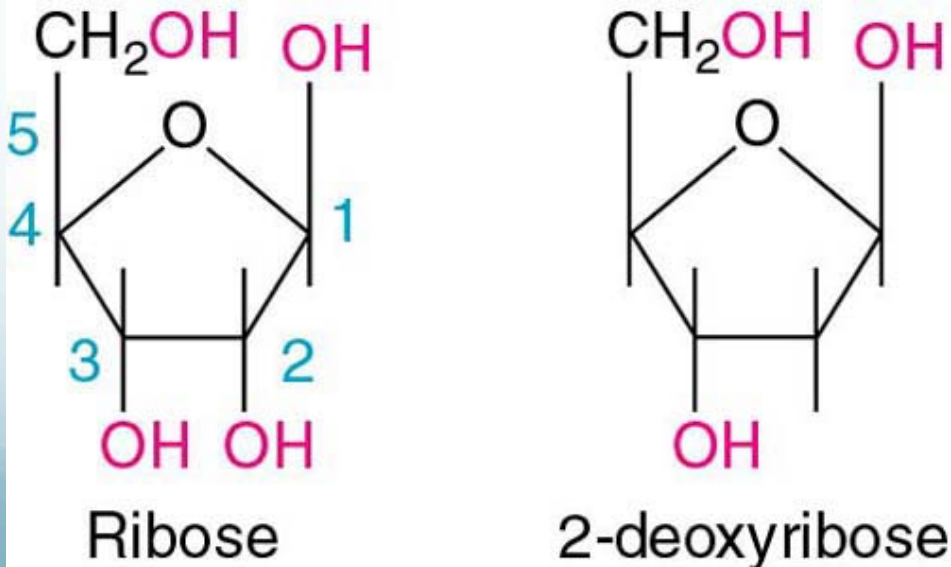
1-nitrogen base (purin & pyrimidine)

- **2-Purines:** Is characterized by a Five –member associated with a six- member ring Members of purine family are nitrogenous bases named
- 1- Adenine (A)
- 2-Guanine (G)



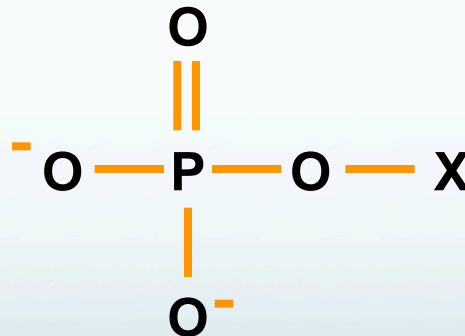
2-sugar (pentose).

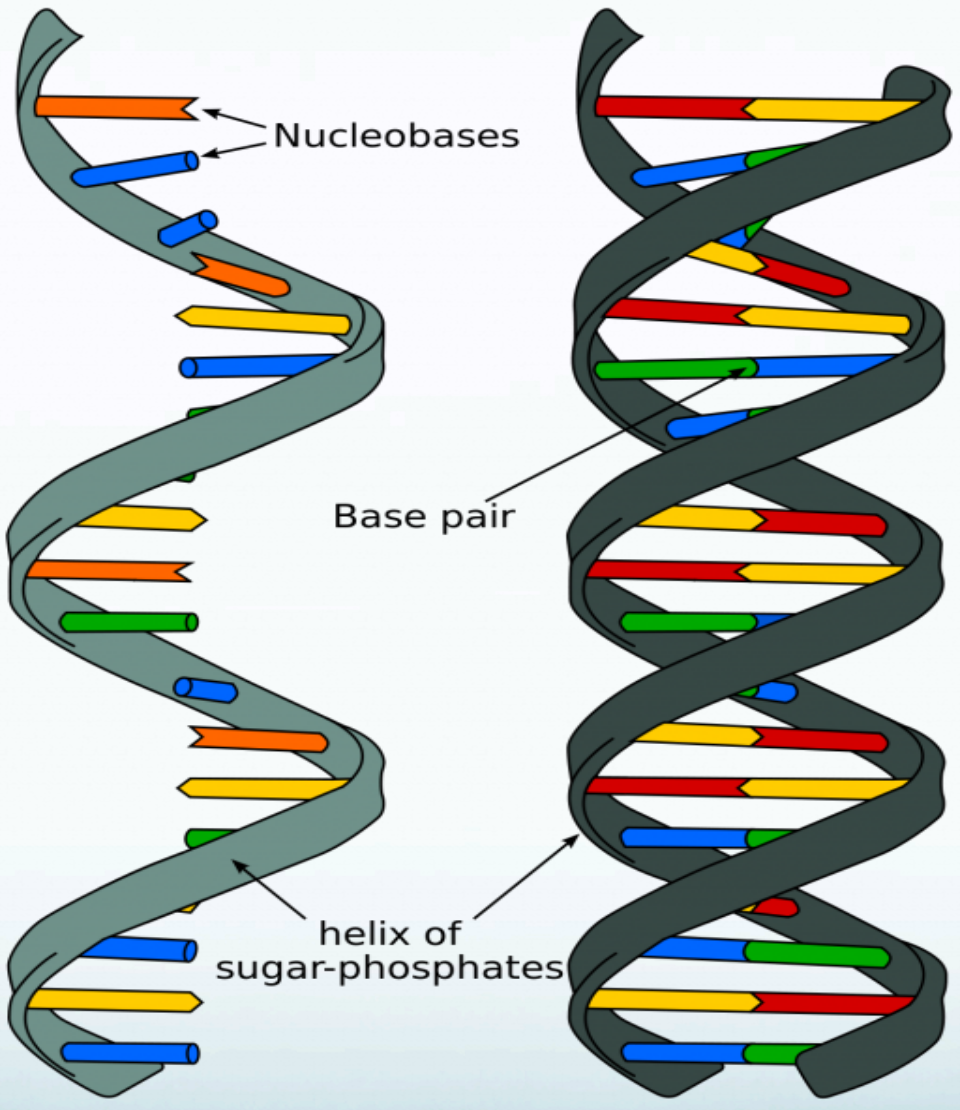
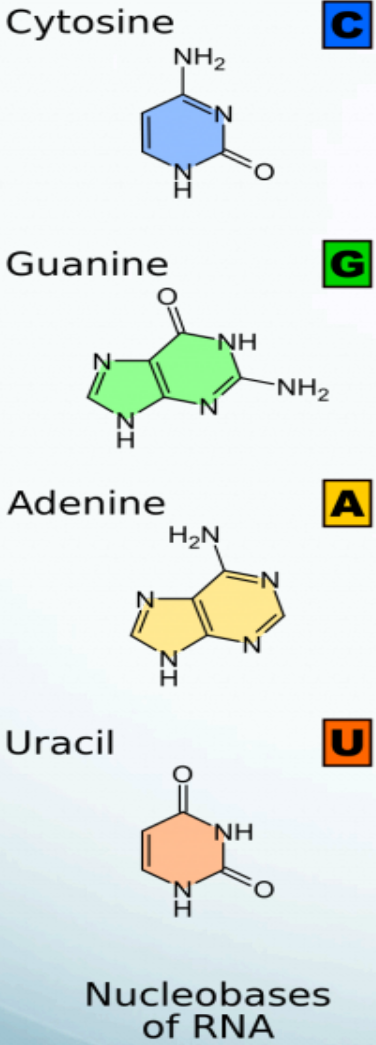
five carbon sugar (**Pentose sugar**) either ribose in RNA or deoxyribose in DNA, which in turn is bounded to a **Phosphate group** which attached to carbon number five of the sugar.



Phosphate Group

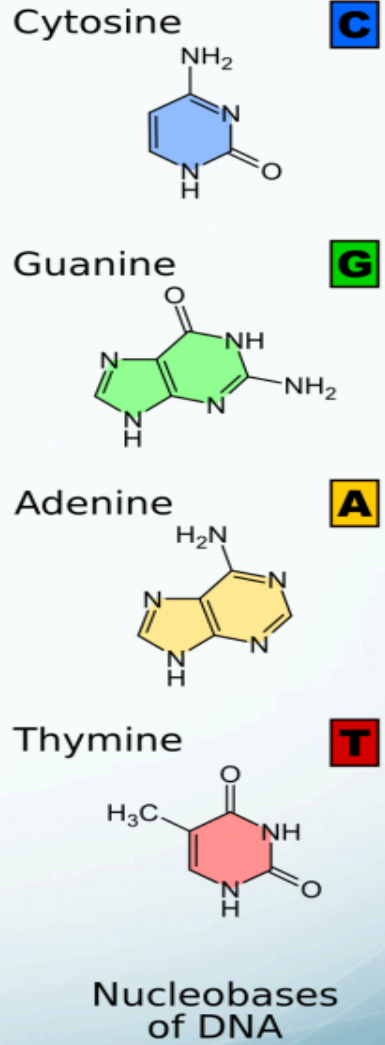
- **Negatively charged** at neutral pH
- Typically attached to **5'** position
- **Nucleic acids contain one phosphate moiety per nucleotide**





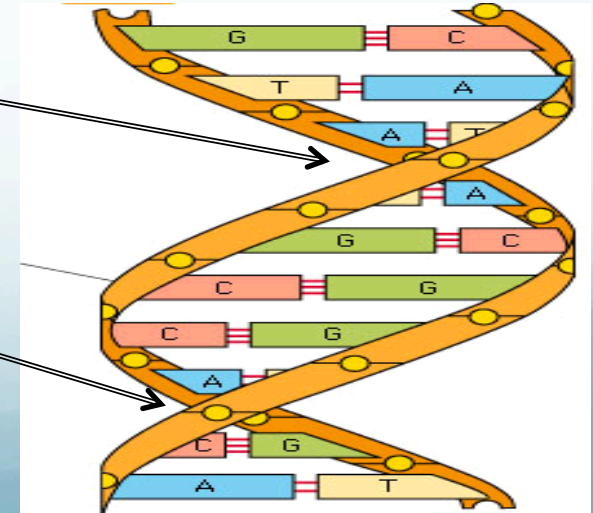
RNA
Ribonucleic acid

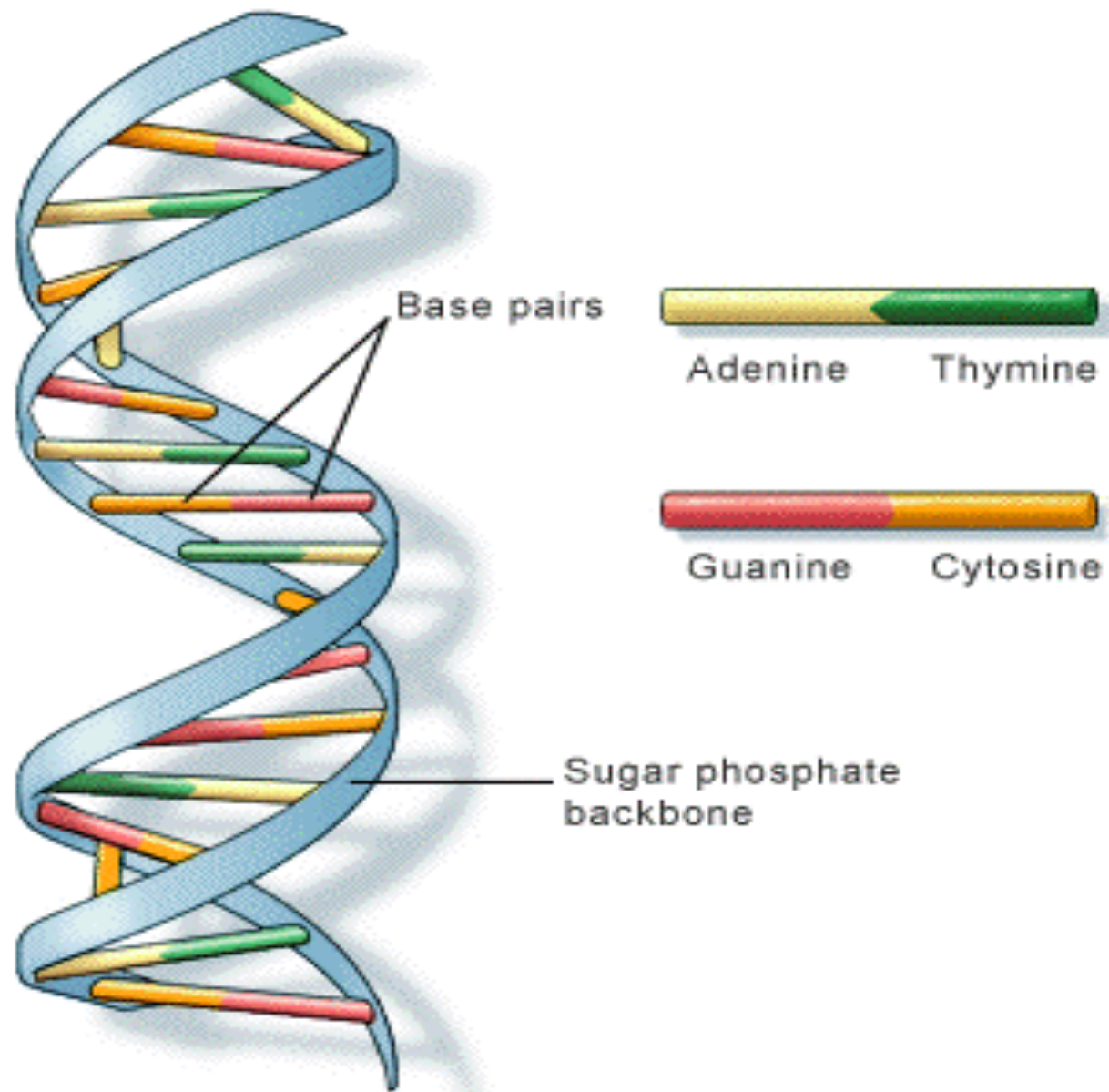
DNA
Deoxyribonucleic acid



Deoxyribonucleic acid (DNA)

- DNA is the genetic material of the cell, is located in the nucleus, where it acts as a template for RNA transcription .
- DNA is composed of two right-handed helical polynucleotide chains or strands are held together by hydrogen bonds between the paired bases.
- Most DNA molecules are very long, one long DNA molecule represent a large number of genes, each one is a particular segment of the double helix.
- DNA structure consists of “major” grooves
- and “minor” grooves
- Major grooves are critical for binding
- proteins that regulate DNA function





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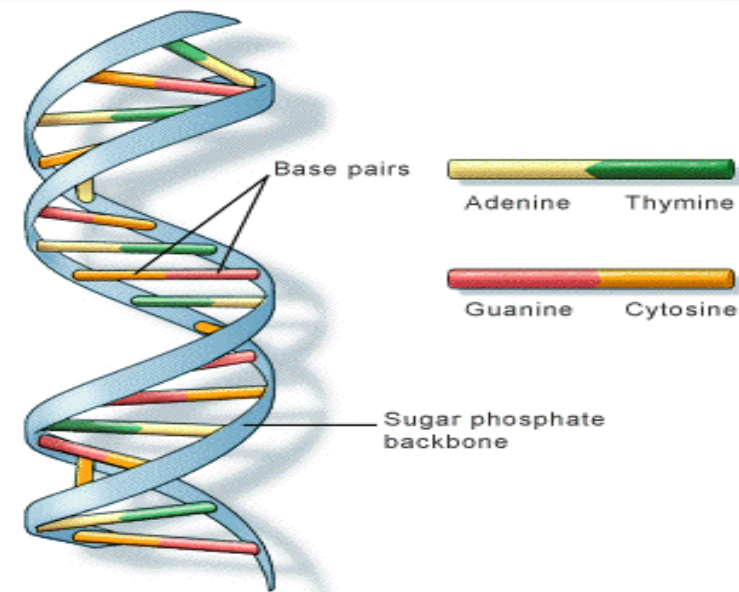
both types of nitrogen bases are found in DNA molecule:

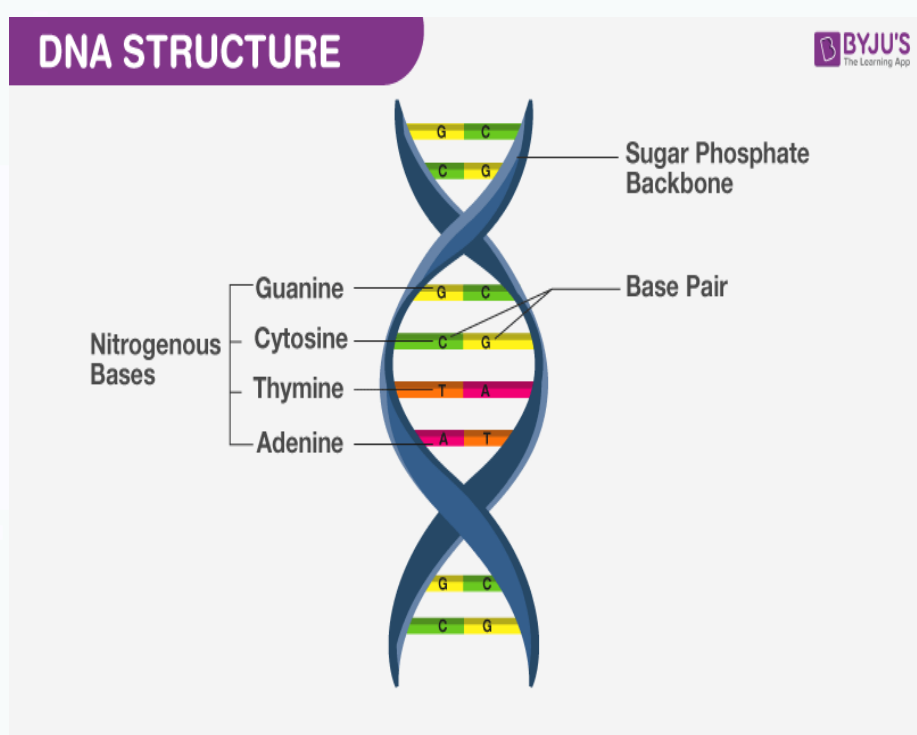
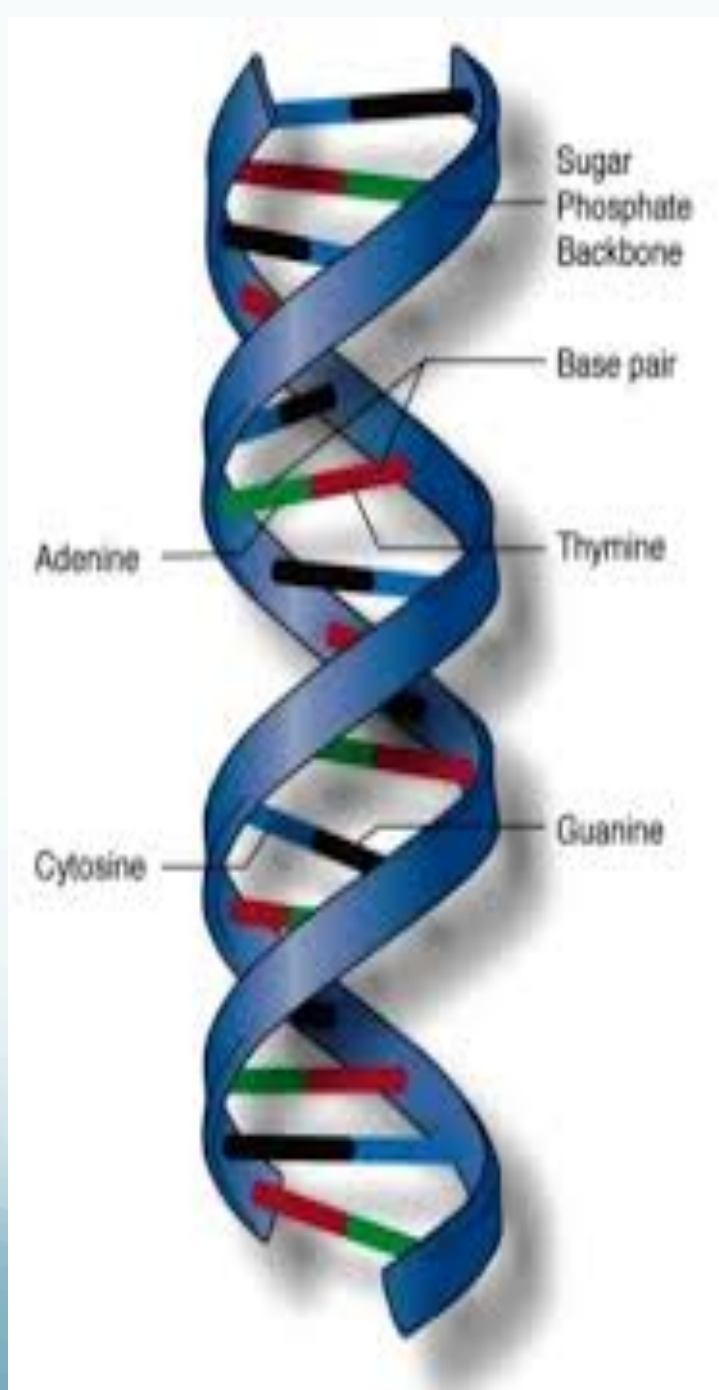
Purine(Adenine & Guanine)

Pyrimidens (Cytosine &Thymine)

Only certain bases are found together in the doublehelix
Adenine (A) always paired with Thymine (T) & Guanine (G)
With Cytosine (C) .

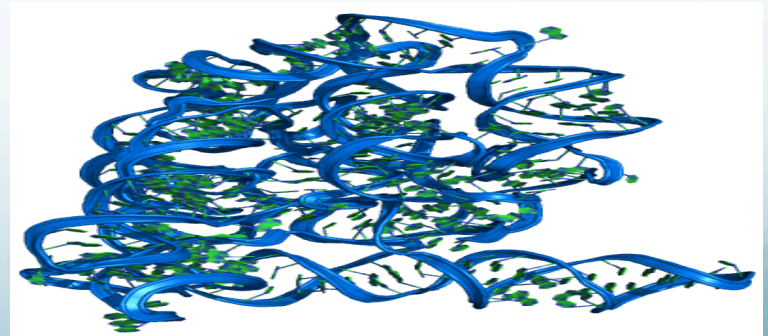
DNA is found in the nuclei of the cells but small amount is
also present in mitochondria.

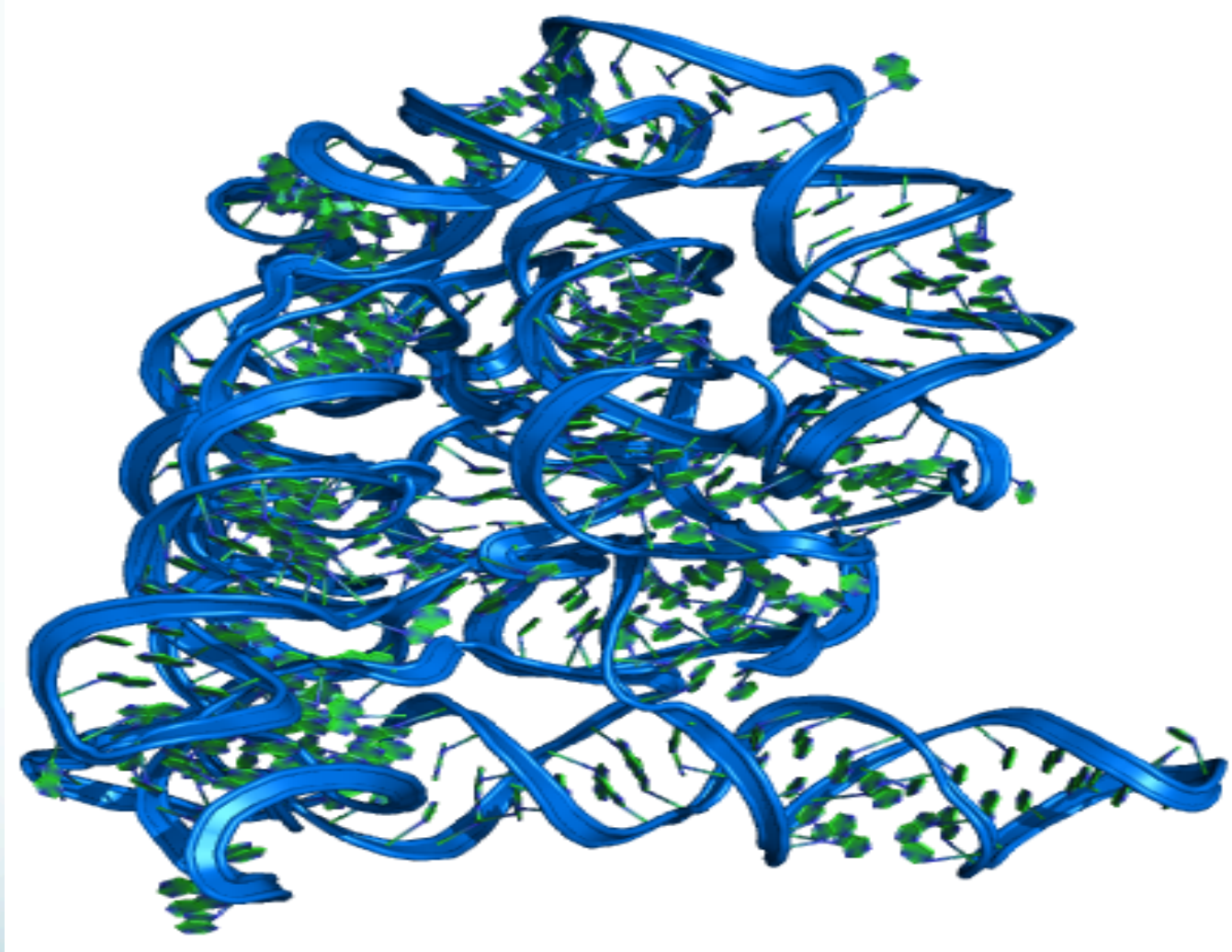




Ribonucleic acid (RNA)

- RNA is similar to DNA except that it is a single-stranded and one of its bases is Uracil instead of Thymine also its sugar is ribose instead of deoxy ribose.
- About %90 of the RNA is present in the cytoplasm of the cell while the remaining %10 is found in the nucleolus.
- RNA is the end product of transcription and three types of RNA are synthesized.





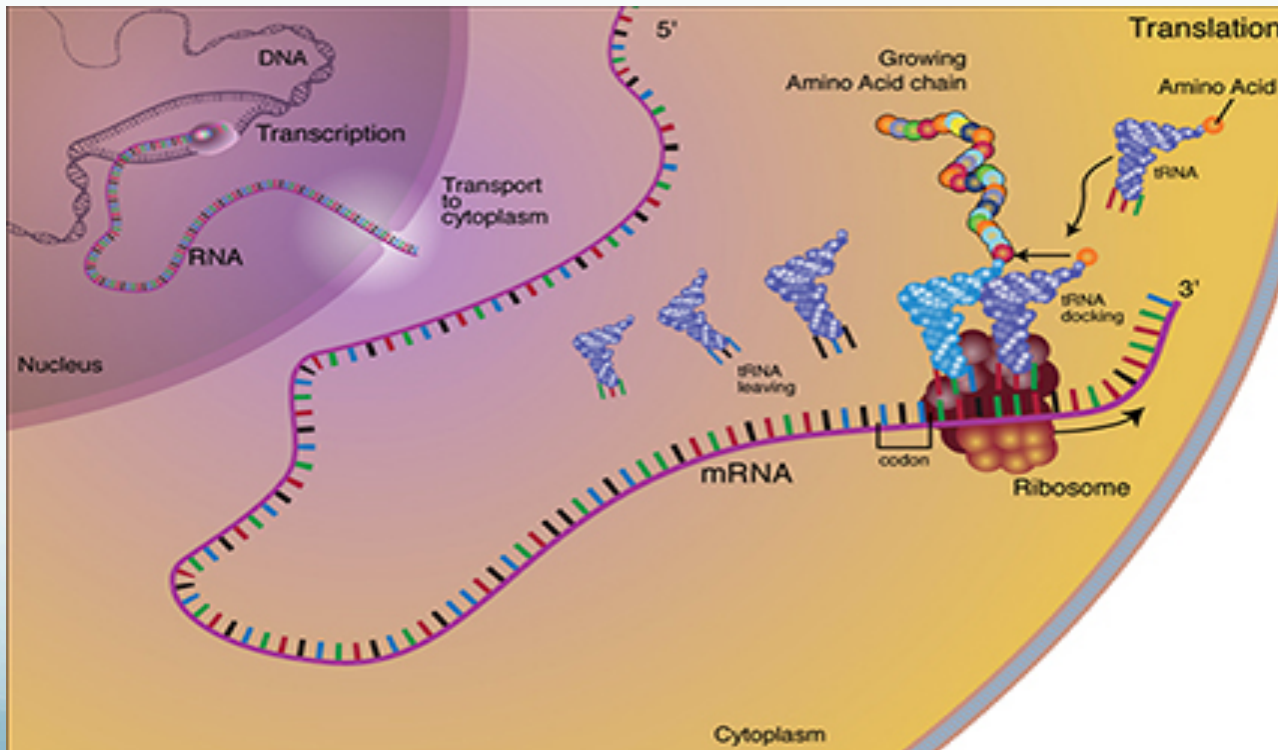
RNA - Molecule

Different types of RNA

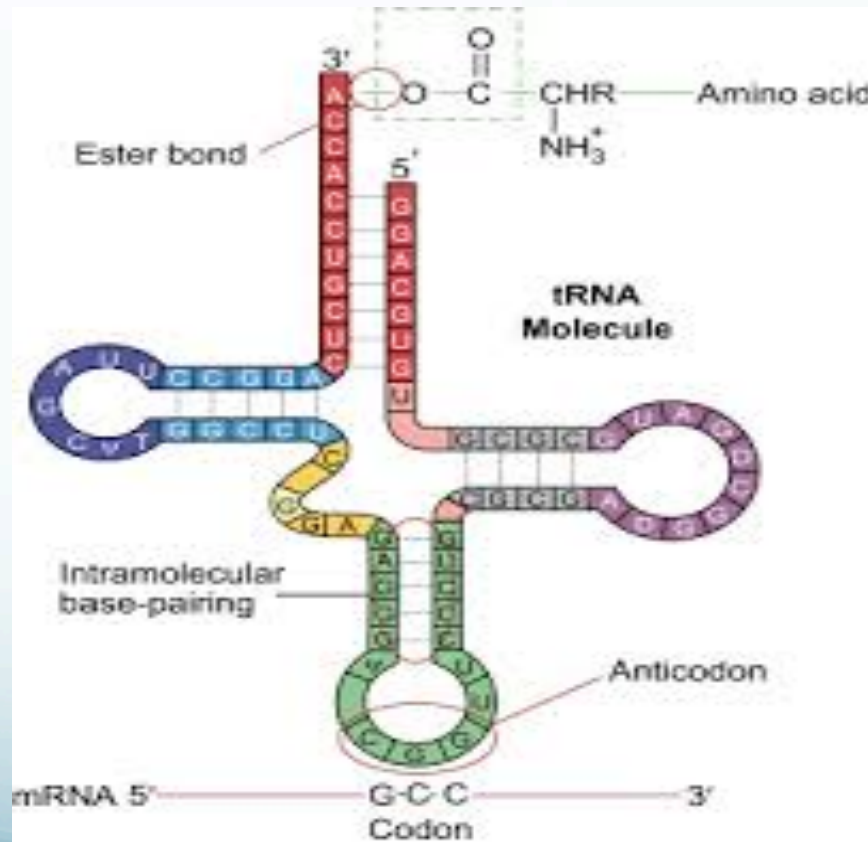
- 1-Messenger RNA (mRNA).
- 2-Transfer RNA (tRNA).
- 3-Ribosomal RNA (rRNA).
-

1-mRNA

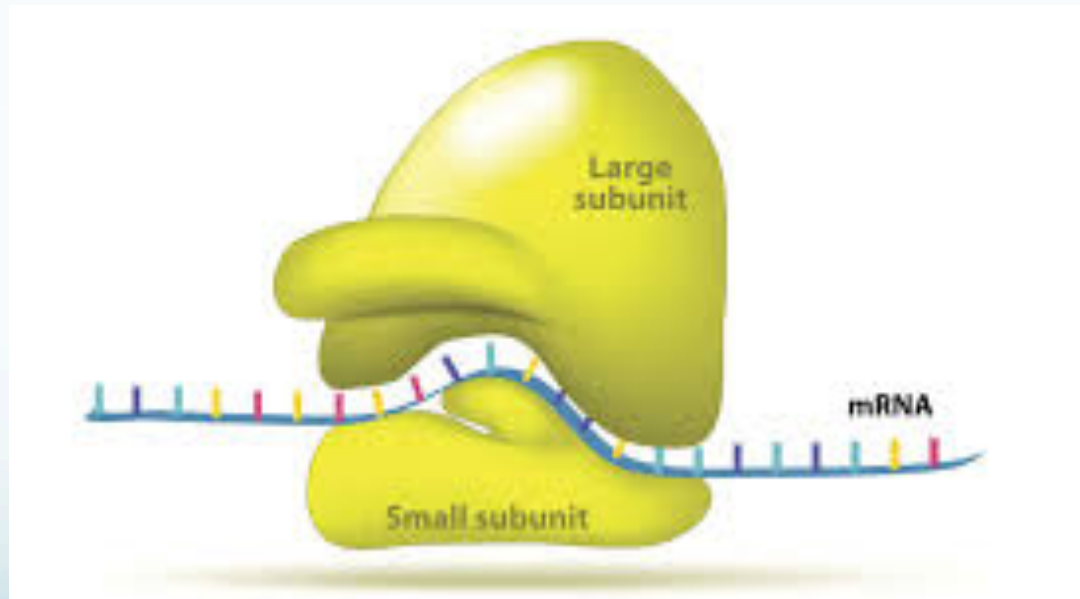
Messenger RNA carries the genetic code from the nucleus To the cytoplasm to act as a template for protein synthesis .

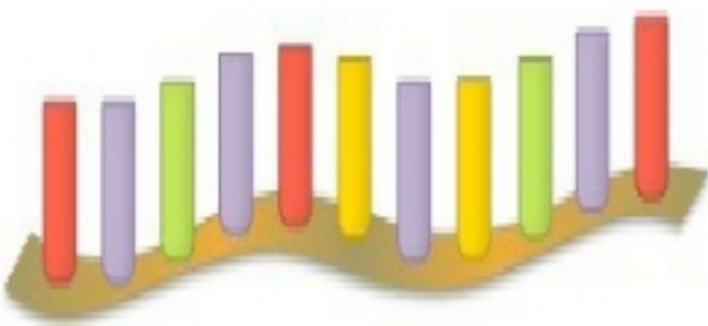


- 2-tRNA:
- Like mRNA transfer RNA picketed up activated amino acids in the cytoplasm to the ribosomal RNA resulting the formation of proteins.

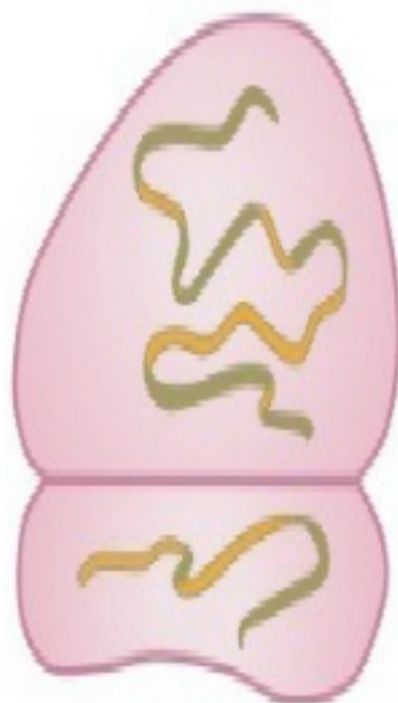


- 3-rRNA :
- Ribosomal RNA are an integral and permanent part of the Ribosome structure (60-80)% of the total cell RNA is rRNA.





Messenger RNA (mRNA)



Ribosomal RNA (rRNA)



Transfer RNA (tRNA)

DNA

RNA

Structure	A double stranded molecule arranged in a helix structure. Each strand is a polymer of subunits - called nucleotides. Each nucleotide contains a phosphate, a 5-carbon deoxyribose sugar, and a nitrogenous base. DNA bases are: - Adenine (A) - Thymine (T) - Cytosine (C) - Guanine (G)	A single stranded molecule and polymer of subunits called nucleotides. Each strand is a polymer of subunits - called nucleotides. Each nucleotide contains a phosphate, a 5-carbon ribose sugar, and a nitrogenous base. RNA bases are: - Adenine (A) - Uracil (U) - Cytosine (C) - Guanine (G)
Function in Humans	Stores genetic information. Located in the cells' nucleus.	Translates genetic information into proteins Located outside the cells' nucleus, in the cytoplasm.

Size

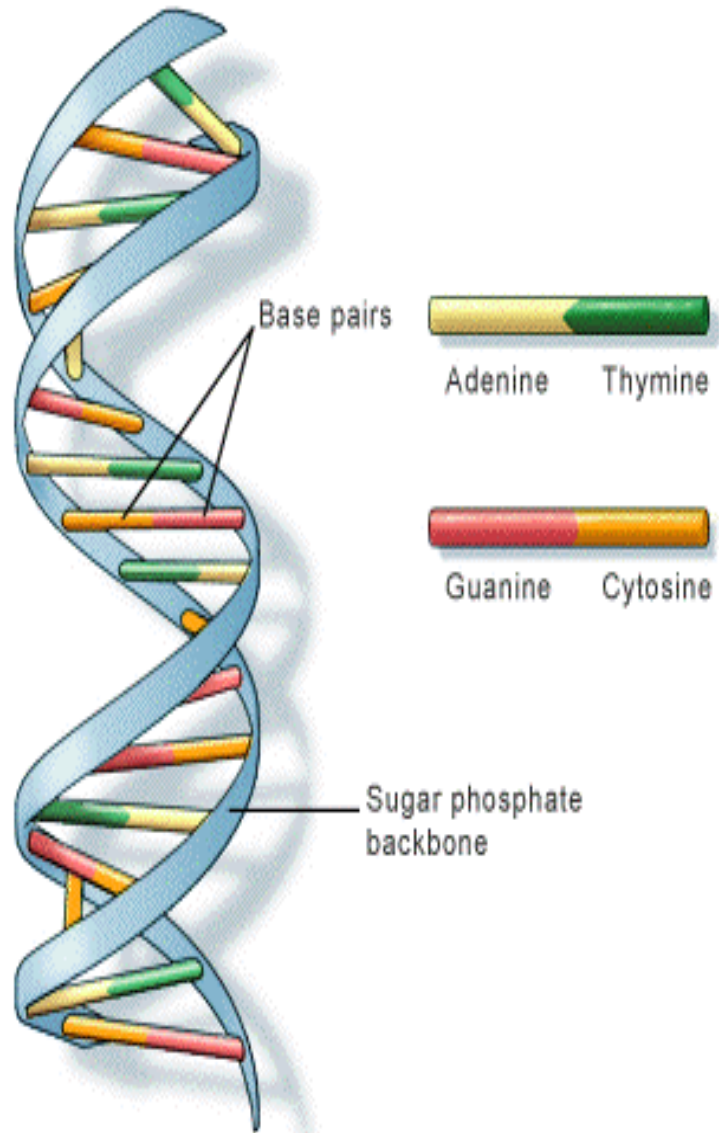
Large polymers

Variable in size but smaller than potential length of DNA polymers

Stability

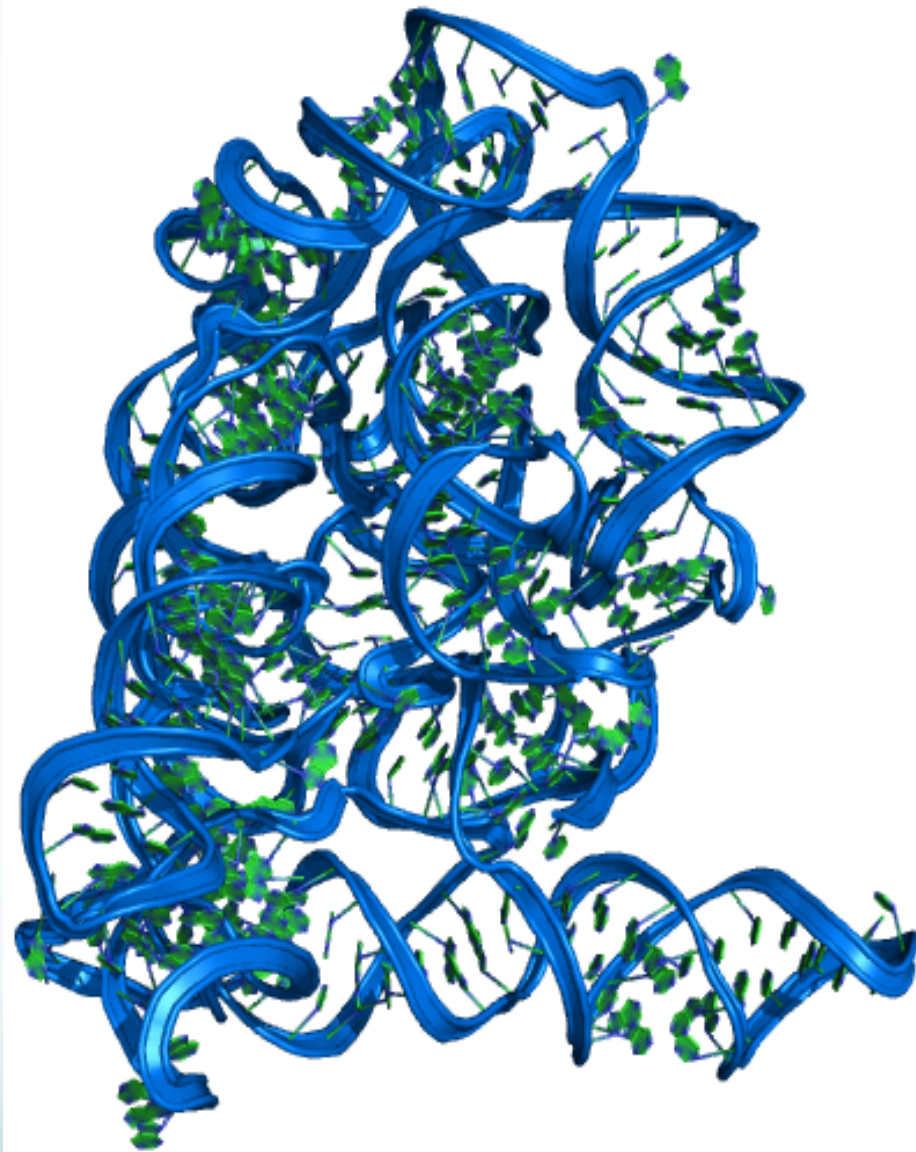
Stable

Unstable



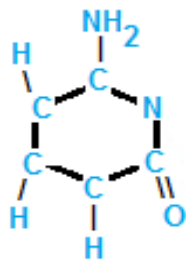
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DNA

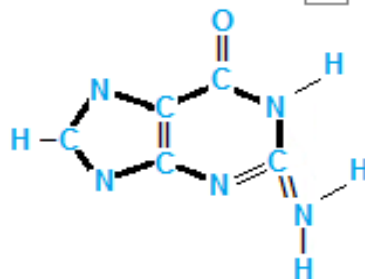


RNA

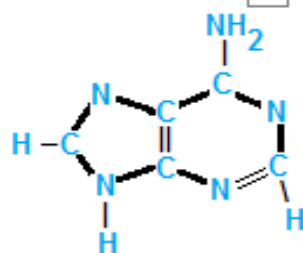
Cytosine [C]



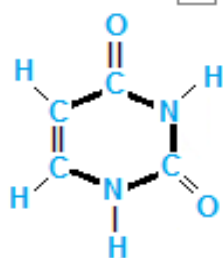
Guanine [G]



Adenine [A]



Uracil [U]



Replace Thymine in RNA

Nitrogenous
Bases

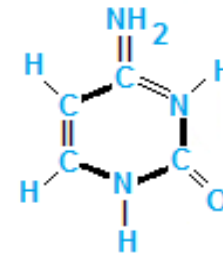


RNA

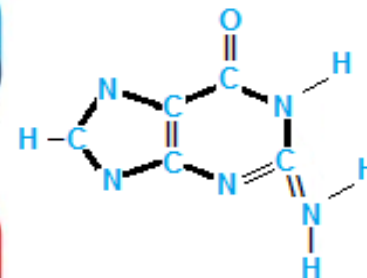


DNA

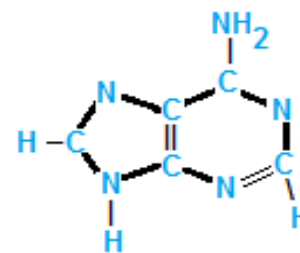
Cytosine [C]



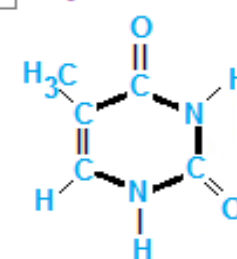
Guanine [G]



Adenine [A]



Thymine [T]



Nitrogenous
Bases



Thank
You

- What are the most important inorganic substances in the cell?
- What is the important of some of them(inorganic substances)
- What is the most important organic components of the cell?
- Why they named as organic components?
- Classify carbohydrate
- Classify polysaccarides according to function and structure.
- What is protein? The building block of different types of proteins are.....
- Classify proteins according to there function in human
- lipids are grouped together in one group according to.....

- What are the advantages and dis-advantages of different types of lipid in human body.
 - What is nucleic acid?
 - What is its important?
 - What is the building block of both types ?
- what are the main differences between RNA & DNA